

RESEARCH ARTICLE

Temporal Heterogeneous Graph Contrastive Learning for Fraud Detection in Credit Card Transactions

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ABSTRACT Credit card fraud detection remains a critical challenge in financial security, characterized by evolving fraud patterns, sparse labeled data, and severe class imbalance. Traditional methods often fail to capture the complex temporal dynamics and heterogeneous relationships inherent in financial transaction networks. To address these limitations, we propose TH-GCL (Temporal Heterogeneous Graph Contrastive Learning), a novel framework that integrates heterogeneous graph modeling, temporal pattern recognition, and contrastive learning for enhanced fraud detection. Our approach constructs a temporal heterogeneous graph incorporating multiple entity types including users, transactions, merchants, and devices, with time-aware edge weights to capture evolving behavioral patterns. We design a temporal-aware graph neural network architecture that learns hierarchical representations by jointly modeling structural dependencies and temporal evolution patterns. Furthermore, we introduce a dual-view contrastive learning mechanism that creates augmented graph views through both structural perturbation and temporal masking, enabling the model to learn robust representations under different perspectives. The contrastive objective encourages the model to distinguish between normal and fraudulent transaction patterns while maintaining consistency across augmented views. Extensive experiments on the IEEE-CIS Fraud Detection dataset demonstrate that TH-GCL achieves superior performance compared to state-of-the-art baselines, with improvements of 5.2% in AUC-ROC and 8.7% in AUC-PR. Ablation studies confirm the effectiveness of each component, while analysis of learned representations reveals meaningful fraud pattern discovery. Our framework exhibits strong generalization capability across different time periods and transaction volumes, making it practical for real-world deployment in dynamic fraud detection scenarios.

INDEX TERMS Fraud detection, heterogeneous graph neural networks, temporal modeling, contrastive learning.

I. INTRODUCTION

Credit card fraud has emerged as one of the most pervasive threats to the global financial ecosystem, with losses exceeding \$32 billion annually worldwide [1]. The rapid digitization of payment systems and the proliferation of e-commerce platforms have created unprecedented opportunities for

fraudulent activities, necessitating sophisticated detection mechanisms that can adapt to evolving threat landscapes [2], [3]. Traditional rule-based systems and conventional machine learning approaches, while foundational, struggle to keep pace with the sophisticated tactics employed by modern fraudsters who continuously adapt their strategies to circumvent detection systems [4].

The inherent challenges in credit card fraud detection are multifaceted and interconnected. First, the extreme

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class imbalance problem, where fraudulent transactions typically constitute less than 1% of all transactions [5], [6], creates significant difficulties for conventional supervised learning algorithms. Second, the temporal nature of fraud patterns introduces concept drift, where the characteristics of fraudulent behavior evolve continuously, rendering static models ineffective over time [7]. Third, the heterogeneous nature of transaction ecosystems involves multiple entity types including cardholders, merchants, payment processors, and devices, each contributing distinct behavioral signals that are challenging to model cohesively [8].

Recent advances in graph neural networks (GNNs) have demonstrated remarkable potential in fraud detection by naturally modeling the relational structure inherent in financial networks [9], [10]. These approaches represent transactions, users, and merchants as nodes in a graph, with edges capturing various relationships such as shared payment methods, geographical proximity, or temporal co-occurrence [11], [12]. However, existing graph-based methods face several critical limitations. Most approaches treat the graph as static, failing to capture the temporal evolution of fraud patterns [13]. Additionally, they often struggle with the heterogeneous nature of financial entities, typically reducing all nodes to homogeneous representations that lose important type-specific information [14]. The emergence of contrastive learning as a powerful representation learning paradigm has shown promise in addressing data scarcity and class imbalance issues across various domains [15], [16]. In the context of fraud detection, contrastive learning can potentially leverage unlabeled transaction data to learn meaningful representations that distinguish between normal and anomalous patterns [17]. However, directly applying existing contrastive learning frameworks to fraud detection presents unique challenges due to the temporal and heterogeneous nature of financial data.

To address these limitations, we propose TH-GCL (Temporal Heterogeneous Graph Contrastive Learning), a novel framework that synergistically combines heterogeneous graph modeling, temporal pattern recognition, and contrastive learning for enhanced fraud detection. Our approach constructs temporal heterogeneous graphs that explicitly model multiple entity types including users, transactions, merchants, and devices with time-aware edge weights, naturally capturing both structural dependencies and temporal evolution patterns inherent in transaction networks. We develop a temporal-aware graph neural network that incorporates temporal attention mechanisms and hierarchical message passing to learn rich representations capturing both local neighborhood structures and global temporal trends. Furthermore, we introduce an innovative dual-view contrastive learning mechanism that creates augmented graph views through both structural perturbation and temporal masking. This approach enables the model to learn robust representations that are invariant to noise while being sensitive to fraud-relevant patterns. Through extensive experiments on the IEEE-CIS Fraud Detection dataset, we demonstrate

significant improvements over state-of-the-art baselines, with comprehensive analysis including ablation studies, robustness evaluation, and interpretability insights that validate the effectiveness of our framework.

The remainder of this paper is organized as follows. Section II reviews related work in fraud detection, graph neural networks, and contrastive learning. Section III provides necessary preliminaries and problem formulation. Section IV presents our TH-GCL methodology in detail. Section V describes our experimental setup and results. Finally, Section VI concludes the paper and discusses future research directions.

II. RELATED WORK

A. TRADITIONAL FRAUD DETECTION METHODS

Early fraud detection systems primarily relied on rule-based approaches and statistical methods. Rule-based systems utilize predefined thresholds and business logic to identify suspicious transactions, but suffer from high false positive rates and inability to adapt to evolving fraud patterns [18]. Statistical approaches such as logistic regression and decision trees have been widely adopted due to their interpretability, but struggle with the complex non-linear relationships present in modern fraud scenarios [19]. The introduction of ensemble methods marked a significant advancement in fraud detection capabilities. Random Forest and Gradient Boosting Machine (GBM) approaches demonstrated improved performance by combining multiple weak learners [20]. AdaBoost and XGBoost further enhanced detection accuracy through adaptive boosting mechanisms [21]. However, these methods still face limitations in handling temporal dependencies and the complex relational structures inherent in transaction networks. Deep learning approaches have gained prominence in recent years, with neural networks showing superior performance in capturing non-linear patterns. Deep Neural Networks (DNNs) and Convolutional Neural Networks (CNNs) have been applied to fraud detection with notable success [22], [23]. Recurrent Neural Networks (RNNs) and Long Short-Term Memory (LSTM) networks have demonstrated effectiveness in modeling sequential transaction patterns [24]. Despite these advances, traditional deep learning methods treat transactions as independent instances, failing to leverage the rich relational information present in financial ecosystems.

B. GRAPH-BASED FRAUD DETECTION

The recognition that fraud detection inherently involves relational data has led to the emergence of graph-based approaches. Early graph-based methods focused on constructing homogeneous graphs where all nodes represent the same entity type, typically transactions or users [8]. These approaches apply traditional graph mining techniques such as community detection and centrality analysis to identify anomalous patterns [25]. Graph Neural Networks (GNNs) have revolutionized graph-based fraud detection by

enabling end-to-end learning on graph-structured data. Graph Convolutional Networks (GCNs) aggregate neighborhood information through message passing, allowing nodes to learn representations that capture both local and global graph structure [26]. GraphSAGE extends this approach by sampling and aggregating from node neighborhoods, improving scalability for large graphs [27]. Graph Attention Networks (GATs) introduce attention mechanisms to weight neighbor contributions, enabling more sophisticated aggregation strategies [28]. Recent work has focused on adapting GNNs specifically for fraud detection scenarios. FdGars proposes a GCN-based approach with attention mechanisms for financial fraud detection [9]. CARE-GNN addresses the challenge of camouflaged fraudsters by incorporating reinforcement learning for neighbor selection [11]. FraudGNN-RL tackles sophisticated financial fraud detection by integrating GNNs with reinforcement learning for dynamic threshold optimization and federated learning for privacy-preserving multi-institutional collaboration [29]. GEM utilizes multi-view learning to capture different types of relationships in financial networks [30]. However, most existing approaches treat graphs as static structures, failing to capture the temporal evolution of fraud patterns.

C. HETEROGENEOUS GRAPH NEURAL NETWORKS

The limitation of homogeneous graph representations has motivated the development of heterogeneous graph neural networks that can handle multiple node and edge types simultaneously [31]. Metapath2vec learns node embeddings by performing random walks along predefined meta-paths in heterogeneous graphs [32]. HAN (Heterogeneous Attention Network) introduces hierarchical attention mechanisms to aggregate information from different types of neighbors and meta-paths [33]. More recent approaches have focused on automatic meta-path discovery and learning. HGT (Heterogeneous Graph Transformer) uses transformer architecture with heterogeneous mutual attention to capture complex dependencies [34]. MAGNN incorporates intra-metapath and inter-metapath aggregation to learn comprehensive node representations [35]. In the fraud detection domain, Hetero-SAGA proposes a heterogeneous graph approach for anti-money laundering [36], while FI-GRL develops a heterogeneous graph representation learning framework for financial intelligence [37]. Despite these advances, most heterogeneous GNN approaches still treat graphs as static snapshots, missing the crucial temporal dimension that characterizes evolving fraud patterns. The integration of temporal modeling with heterogeneous graph representation remains an underexplored area with significant potential for fraud detection applications.

D. TEMPORAL GRAPH NEURAL NETWORKS

The importance of temporal information in graph learning has led to the development of temporal graph neural networks [38]. EvolveGCN proposes to evolve GCN parameters

over time using RNNs [39]. DyGCN incorporates temporal convolution operations to capture dynamic graph patterns [40]. TGCN combines graph convolution with gated recurrent units for temporal graph learning [41]. Recent work has focused on continuous-time dynamic graphs where edges have timestamps. TGAT (Temporal Graph Attention Network) uses temporal attention mechanisms to aggregate information from temporal neighbors [42]. TGN (Temporal Graph Networks) maintains memory modules for nodes to capture long-term dependencies [43]. APAN introduces adaptive temporal attention for dynamic graph learning [44]. However, most temporal GNN approaches focus on homogeneous graphs and do not address the heterogeneous nature of financial transaction networks. The combination of temporal modeling with heterogeneous graph representation for fraud detection remains largely unexplored, representing a significant gap that our work aims to address.

E. CONTRASTIVE LEARNING

Contrastive learning has emerged as a powerful paradigm for representation learning, particularly in scenarios with limited labeled data [45]. SimCLR demonstrates the effectiveness of contrastive learning for visual representation learning through data augmentation and contrastive objectives [15]. MoCo introduces momentum-based contrastive learning for efficient training [16]. BYOL shows that contrastive learning can work without negative samples through asymmetric networks [46]. In the graph domain, contrastive learning has shown promise for learning robust graph representations. DGI maximizes mutual information between node representations and graph summaries [47]. GraphCL applies various graph augmentation strategies for contrastive learning [48]. GCC proposes graph contrastive coding for pre-training graph neural networks [17]. GRACE introduces adaptive augmentation strategies for graph contrastive learning [49]. Recent work has begun exploring contrastive learning for fraud detection. FdGars incorporates attention-based contrastive learning for financial fraud detection [9]. However, existing approaches do not specifically address the challenges of temporal heterogeneous graphs in fraud detection scenarios. The integration of contrastive learning with temporal heterogeneous graph modeling represents a promising but underexplored direction that our work investigates.

III. PRELIMINARIES

A. HETEROGENEOUS GRAPH

A heterogeneous graph is defined as $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{A}, \mathcal{R})$, where $\mathcal{V}$ is the set of nodes, $\mathcal{E}$ is the set of edges, $\mathcal{A}$ is the node type mapping function $\phi : \mathcal{V} \rightarrow \mathcal{A}$, and $\mathcal{R}$ is the edge type mapping function $\psi : \mathcal{E} \rightarrow \mathcal{R}$. A heterogeneous graph satisfies $|\mathcal{A}| + |\mathcal{R}| > 2$, meaning there are multiple node types or edge types. In the context of fraud detection, we consider four main node types: users ($\mathcal{V}_u$), transactions ($\mathcal{V}_t$), merchants ($\mathcal{V}_m$), and devices ($\mathcal{V}_d$). The edge types include user-transaction relationships, transaction-merchant

relationships, and device-transaction relationships, among others.

B. TEMPORAL HETEROGENEOUS GRAPH

A temporal heterogeneous graph extends the heterogeneous graph definition by incorporating temporal information. We define it as $\mathcal{G}^T = (\mathcal{V}, \mathcal{E}, \mathcal{A}, \mathcal{R}, T)$, where T represents the temporal dimension. Each edge $e_{ij} \in \mathcal{E}$ is associated with a timestamp t_{ij} , indicating when the relationship was established or when the interaction occurred. For fraud detection, temporal information is crucial as it captures the evolution of transaction patterns and user behaviors over time. We denote the temporal heterogeneous graph at time t as $\mathcal{G}^t$, which includes all nodes and edges with timestamps up to time t .

C. GRAPH NEURAL NETWORKS

Graph Neural Networks operate by iteratively aggregating information from node neighborhoods. For a node v_i , the message passing process can be formulated as:

$$\mathbf{m}_{v_i}^{(l+1)} = \text{AGGREGATE}^{(l)} \left(\{ \mathbf{h}_{v_j}^{(l)} : v_j \in \mathcal{N}(v_i) \} \right) \quad (1)$$

$$\mathbf{h}_{v_i}^{(l+1)} = \text{UPDATE}^{(l)} \left(\mathbf{h}_{v_i}^{(l)}, \mathbf{m}_{v_i}^{(l+1)} \right) \quad (2)$$

where $\mathbf{h}_{v_i}^{(l)}$ denotes the representation of node v_i at layer l , $\mathcal{N}(v_i)$ represents the neighbors of node v_i , and AGGREGATE and UPDATE are learnable functions.

D. CONTRASTIVE LEARNING

Contrastive learning aims to learn representations by maximizing agreement between positive pairs while minimizing agreement between negative pairs. The contrastive loss is typically formulated using the InfoNCE loss:

$$\mathcal{L}_{\text{contrast}} = -\log \frac{\exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_i^+)/\tau)}{\sum_{k=1}^N \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_k)/\tau)} \quad (3)$$

where $\mathbf{z}_i$ and $\mathbf{z}_i^+$ form a positive pair, $\mathbf{z}_k$ represents negative samples, $\text{sim}(\cdot, \cdot)$ is a similarity function (typically cosine similarity), and τ is a temperature parameter.

E. PROBLEM FORMULATION

Given a temporal heterogeneous graph $\mathcal{G}^T$ representing financial transaction networks, our goal is to detect fraudulent transactions. Formally, we aim to learn a function $f : \mathcal{V}_t \rightarrow \{0, 1\}$ that maps each transaction node to a binary label, where 1 indicates fraud and 0 indicates a legitimate transaction.

The key challenges in this problem include: (1) **Extreme class imbalance**, where fraudulent transactions constitute less than 1% of all transactions; (2) **Temporal concept drift**, where fraud patterns evolve continuously over time; (3) **Heterogeneous relationships**, involving multiple entity types with complex interactions; and (4) **Limited labeled data**, as obtaining fraud labels is expensive and time-consuming.

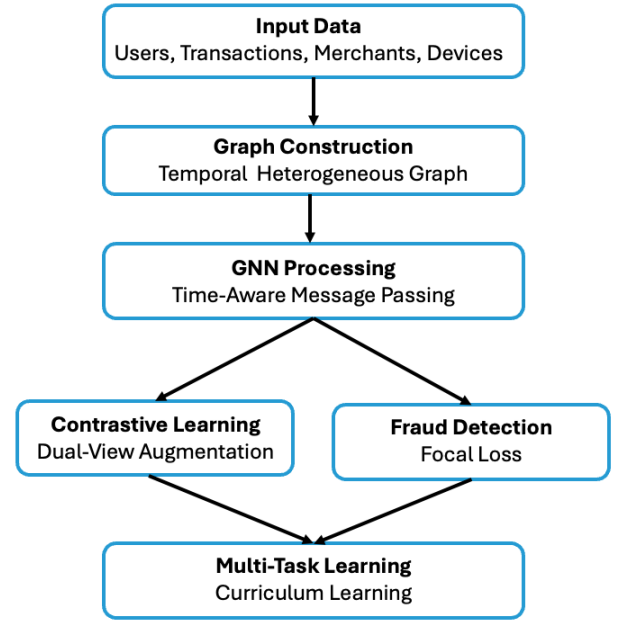


FIGURE 1. The flowchart of our TH-GCL framework.

Our objective is to develop a framework that can effectively leverage the temporal and heterogeneous structure of transaction networks while addressing the challenges of sparse labels and class imbalance through contrastive learning. Specifically, we aim to learn robust node representations $\mathbf{h}_v$ for each transaction node $v \in \mathcal{V}_t$ that can accurately distinguish between fraudulent and legitimate transactions, even under limited supervision.

IV. METHODOLOGY

In this section, we present TH-GCL (Temporal Heterogeneous Graph Contrastive Learning), a comprehensive framework that addresses the key challenges in credit card fraud detection through three synergistic components: temporal heterogeneous graph construction, time-aware graph neural architecture, and dual-view contrastive learning. The whole flowchart has been provided in Figure 1.

A. TEMPORAL HETEROGENEOUS GRAPH CONSTRUCTION

1) GRAPH SCHEMA DESIGN

The foundation of our approach lies in constructing a comprehensive temporal heterogeneous graph that captures the multi-faceted nature of financial transaction ecosystems. We design a graph schema that incorporates four primary node types and their temporal interactions.

a: NODE TYPES

Our heterogeneous graph includes users ($\mathcal{V}_u$), transactions ($\mathcal{V}_t$), merchants ($\mathcal{V}_m$), and devices ($\mathcal{V}_d$). Each node type captures different aspects of the transaction ecosystem. User nodes represent cardholders with features including demographic information, account history, and behavioral patterns. Transaction nodes contain transaction-specific

attributes such as amount, location, and timing. Merchant nodes encode business characteristics including category, location, and transaction volume patterns. Device nodes capture device-specific information such as device type, IP address, and usage patterns.

b: EDGE TYPES AND TEMPORAL WEIGHTING

We define multiple edge types to capture various relationships: user-transaction edges ($\mathcal{E}_{ut}$) representing ownership relationships, transaction-merchant edges ($\mathcal{E}_{tm}$) indicating where transactions occur, device-transaction edges ($\mathcal{E}_{dt}$) showing which device initiated the transaction, and user-device edges ($\mathcal{E}_{ud}$) representing device ownership patterns. To incorporate temporal dynamics, we introduce time-decay edge weights that emphasize recent interactions while preserving historical context. For an edge e_{ij} established at time t_{ij} , the temporal weight at current time t is computed as:

$$w_{ij}(t) = \exp\left(-\alpha \cdot \frac{t - t_{ij}}{\Delta t}\right) \quad (4)$$

where α is a decay factor and Δt is a normalization constant. This weighting scheme ensures that recent interactions have stronger influence while maintaining long-term memory.

2) FEATURE ENGINEERING

a: NODE FEATURE CONSTRUCTION

For each node type, we construct comprehensive feature representations that capture both static attributes and dynamic behavioral patterns. User features include demographic information, account age, historical transaction statistics, and behavioral patterns such as spending frequency and amount distributions. Transaction features encompass amount, timestamp, location coordinates, merchant category, and derived features such as time-of-day indicators and weekend flags.

b: TEMPORAL FEATURE AGGREGATION

To capture temporal patterns, we compute time-windowed statistical features for each node. For user nodes, we calculate rolling statistics of transaction amounts, frequencies, and patterns over multiple time windows (e.g., 1 day, 1 week, 1 month). These features are formulated as:

$$f_{u,w}^{(agg)} = \text{AGG}(\{f_{t_i} : t_i \in \mathcal{T}_u^w\}) \quad (5)$$

where $\mathcal{T}_u^w$ represents transactions associated with user u within time window w , and AGG denotes aggregation functions including mean, standard deviation, maximum, and quantiles.

B. TIME-AWARE GRAPH NEURAL ARCHITECTURE

1) HETEROGENEOUS MESSAGE PASSING

Our time-aware graph neural network addresses the challenge of learning representations that capture both heterogeneous relationships and temporal dynamics. The core innovation lies in designing message passing mechanisms that can handle multiple node types while preserving temporal information.

a: TYPE-SPECIFIC MESSAGE COMPUTATION

For heterogeneous message passing, we compute type-specific messages that account for different semantic meanings of various edge types. For a target node v_i of type τ , the message from neighbor v_j of type σ through edge type r is computed as:

$$\mathbf{m}_{ij}^{(l)} = \mathbf{W}_r^{(l)} \cdot \text{ATTENTION}^{(l)}(\mathbf{h}_i^{(l)}, \mathbf{h}_j^{(l)}, r) \cdot \mathbf{h}_j^{(l)} \quad (6)$$

where $\mathbf{W}_r^{(l)}$ is the learnable transformation matrix for edge type r , and ATTENTION computes attention weights based on node representations and edge type.

b: TEMPORAL ATTENTION MECHANISM

To incorporate temporal information, we design a temporal attention mechanism that weights messages based on both semantic relevance and temporal significance:

$$\begin{aligned} & \text{ATTENTION}^{(l)}(\mathbf{h}_i^{(l)}, \mathbf{h}_j^{(l)}, r, t_{ij}) \\ &= \text{softmax}\left(\frac{(\mathbf{W}_q^{(l)} \mathbf{h}_i^{(l)})^T (\mathbf{W}_k^{(l)} \mathbf{h}_j^{(l)})}{\sqrt{d}} \cdot w_{ij}(t)\right) \end{aligned} \quad (7)$$

where $\mathbf{W}_q^{(l)}$ and $\mathbf{W}_k^{(l)}$ are query and key transformation matrices, d is the feature dimension, and $w_{ij}(t)$ is the temporal weight defined in Equation (1).

2) HIERARCHICAL REPRESENTATION LEARNING

a: MULTI-LEVEL AGGREGATION

Our architecture employs hierarchical aggregation to capture information at different granularities. We first aggregate messages within each edge type (intra-metapath aggregation), then combine information across different edge types (inter-metapath aggregation). For intra-metapath aggregation, messages from neighbors of the same type are combined:

$$\mathbf{m}_i^{(l,r)} = \text{AGGREGATE}\left(\{\mathbf{m}_{ij}^{(l)} : v_j \in \mathcal{N}_r(v_i)\}\right) \quad (8)$$

where $\mathcal{N}_r(v_i)$ denotes neighbors of node v_i connected through edge type r . For inter-metapath aggregation, we combine information across different edge types using learned attention weights:

$$\mathbf{m}_i^{(l)} = \sum_{r \in \mathcal{R}} \beta_r^{(l)} \cdot \mathbf{m}_i^{(l,r)} \quad (9)$$

where $\beta_r^{(l)}$ are learned attention weights for edge type r at layer l .

b: NODE REPRESENTATION UPDATE

The final node representation update incorporates both the aggregated messages and the node's current representation through a gated mechanism:

$$\mathbf{h}_i^{(l+1)} = \text{GRU}(\mathbf{h}_i^{(l)}, \mathbf{m}_i^{(l)}) \quad (10)$$

This gated update mechanism helps preserve important historical information while incorporating new neighborhood information.

C. DUAL-VIEW CONTRASTIVE LEARNING

The contrastive learning component addresses the challenges of sparse labels and class imbalance by learning robust representations through self-supervised objectives.

1) GRAPH AUGMENTATION STRATEGIES

a: STRUCTURAL AUGMENTATION

We create structural views of the graph through controlled edge perturbation that preserves the overall graph topology while introducing beneficial noise. Our structural augmentation includes: (1) Edge dropout, where we randomly remove a fraction of edges while ensuring graph connectivity; (2) Node feature masking, where we mask certain dimensions of node features; and (3) Subgraph sampling, where we extract connected subgraphs around transaction nodes.

b: TEMPORAL AUGMENTATION

To create temporal views, we employ time-based augmentation strategies: (1) Temporal masking, where we mask transaction information from certain time periods; (2) Temporal resampling, where we sample different time windows for feature computation; and (3) Temporal permutation, where we slightly perturb timestamps within acceptable ranges.

2) CONTRASTIVE OBJECTIVE

a: POSITIVE PAIR CONSTRUCTION

We construct positive pairs using multiple strategies tailored for fraud detection. For a transaction node v_i , positive examples include: (1) Augmented versions of the same transaction; (2) Transactions from the same user with similar characteristics; (3) Transactions at similar merchants within close time windows.

b: MULTI-VIEW CONTRASTIVE LOSS

Our contrastive objective combines information from both structural and temporal views. For a transaction node v_i , we compute representations under different views and maximize agreement between positive pairs:

$$\mathcal{L}_{\text{contrast}} = -\frac{1}{|\mathcal{V}_t|} \sum_{v_i \in \mathcal{V}_t} \log \frac{\sum_{v_j \in \mathcal{P}(v_i)} \exp(\text{sim}(\mathbf{z}_i^s, \mathbf{z}_j^t)/\tau)}{\sum_{v_k \in \mathcal{V}_t} \exp(\text{sim}(\mathbf{z}_i^s, \mathbf{z}_k^t)/\tau)} \quad (11)$$

where $\mathbf{z}_i^s$ and $\mathbf{z}_i^t$ are representations from structural and temporal views respectively, $\mathcal{P}(v_i)$ denotes positive examples for node v_i , and τ is the temperature parameter.

D. MULTI-TASK LEARNING FRAMEWORK

1) JOINT OPTIMIZATION

Our framework jointly optimizes the fraud detection objective and the contrastive learning objective through a multi-task learning approach. The overall loss function combines supervised classification loss with unsupervised contrastive loss:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{fraud}} + \lambda \cdot \mathcal{L}_{\text{contrast}} \quad (12)$$

where $\mathcal{L}_{\text{fraud}}$ is the binary cross-entropy loss for fraud detection, and λ is a hyperparameter controlling the relative importance of contrastive learning.

a: FRAUD DETECTION LOSS

For labeled transaction nodes, we use focal loss to address class imbalance:

$$\mathcal{L}_{\text{fraud}} = -\frac{1}{|\mathcal{L}|} \sum_{v_i \in \mathcal{L}} \alpha_i (1 - p_i)^\gamma \log(p_i) \quad (13)$$

where $\mathcal{L}$ denotes labeled nodes, p_i is the predicted probability, α_i balances positive and negative examples, and γ focuses learning on hard examples.

2) TRAINING STRATEGY

a: CURRICULUM LEARNING

We employ a curriculum learning strategy that gradually increases the difficulty of contrastive examples. Initially, we use easily distinguishable positive and negative pairs, then progressively introduce more challenging examples as training progresses.

b: DYNAMIC TEMPERATURE SCHEDULING

The temperature parameter τ in the contrastive loss is dynamically adjusted during training. We start with a higher temperature to allow more diverse learning, then gradually decrease it to focus on more discriminative features:

$$\tau_t = \tau_0 \cdot \exp(-\beta \cdot t) \quad (14)$$

where τ_0 is the initial temperature, β is the decay rate, and t is the training step. To better illustrate the overall procedure of algorithm, we provide the pseudocode in Algorithm 1.

This comprehensive methodology integrates temporal modeling, heterogeneous graph representation, and contrastive learning into a unified framework that addresses the key challenges in credit card fraud detection while maintaining computational efficiency and scalability.

V. EXPERIMENTS

A. DATASETS

We conduct comprehensive experiments on two real-world credit card fraud detection datasets to evaluate the effectiveness of our TH-GCL framework.

1) IEEE-CIS FRAUD DETECTION DATASET [50]

This dataset is from the IEEE Computational Intelligence Society fraud detection competition and contains 590,540 transactions collected over 6 months. Each transaction includes 394 features covering transaction details, user information, device characteristics, and derived behavioral patterns. The dataset exhibits extreme class imbalance with only 3.5% fraudulent transactions. We construct the temporal heterogeneous graph by extracting 142,853 unique users, 50,217 merchants, and 89,456 devices from the transaction records.

Algorithm 1 TH-GCL: Temporal Heterogeneous Graph Contrastive Learning

Require: Transaction data D , hyperparameters $\{\alpha, \lambda, \tau_0, \beta\}$
Ensure: Trained model parameters θ

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1: // Phase 1: Temporal Heterogeneous Graph Construction
2:  $\mathcal{G}^T \leftarrow \text{BuildHeterogeneousGraph}(D)$ 
3: for each edge  $e_{ij} \in \mathcal{E}$  do
4:    $w_{ij}(t) \leftarrow \exp(-\alpha \cdot \frac{t-t_{ij}}{\Delta t})$ 
5: end for
6:  $\{\mathbf{h}_v^{(0)}\} \leftarrow \text{FeatureEngineering}(\mathcal{G}^T)$ 
7: // Phase 2: Training Loop
8: for epoch  $e = 1$  to  $E$  do
9:    $\tau_e \leftarrow \tau_0 \cdot \exp(-\beta \cdot e)$ 
10:  // Generate augmented views
11:   $\mathcal{G}^s \leftarrow \text{StructuralAugmentation}(\mathcal{G}^T)$ 
12:   $\mathcal{G}^t \leftarrow \text{TemporalAugmentation}(\mathcal{G}^T)$ 
13:  // Forward pass through time-aware GNN
14:  for layer  $l = 0$  to  $L - 1$  do
15:    for each node  $v_i \in \mathcal{V}$  do
16:       $\mathbf{m}_i^{(l)} \leftarrow \text{MessagePassing}(\mathbf{h}_i^{(l)}, \mathcal{N}(v_i), \{w_{ij}\})$ 
17:       $\mathbf{h}_i^{(l+1)} \leftarrow \text{GRU}(\mathbf{h}_i^{(l)}, \mathbf{m}_i^{(l)})$ 
18:    end for
19:  end for
20:  // Get representations from different views
21:   $\{\mathbf{z}_i^s\} \leftarrow \text{GNN}(\mathcal{G}^s, \theta)$ ,  $\{\mathbf{z}_i^t\} \leftarrow \text{GNN}(\mathcal{G}^t, \theta)$ ,  $\{\mathbf{h}_i\} \leftarrow \text{GNN}(\mathcal{G}^T, \theta)$ 
22:  // Compute losses
23:   $\mathcal{P} \leftarrow \text{ConstructPositivePairs}(\mathcal{G}^T)$ 
24:   $\mathcal{L}_{\text{contrast}} \leftarrow -\frac{1}{|\mathcal{V}|} \sum_{v_i \in \mathcal{V}_i} \log \frac{\sum_{v_j \in \mathcal{P}(v_i)} \exp(\text{sim}(\mathbf{z}_i^s, \mathbf{z}_j^t) / \tau_e)}{\sum_{v_k \in \mathcal{V}_i} \exp(\text{sim}(\mathbf{z}_i^s, \mathbf{z}_k^t) / \tau_e)}$ 
25:   $\mathcal{L}_{\text{fraud}} \leftarrow \frac{1}{|\mathcal{L}|} \sum_{v_i \in \mathcal{L}} \text{FocalLoss}(\text{MLP}(\mathbf{h}_i), y_i)$ 
26:  // Joint optimization
27:   $\mathcal{L}_{\text{total}} \leftarrow \mathcal{L}_{\text{fraud}} + \lambda \cdot \mathcal{L}_{\text{contrast}}$ 
28:   $\theta \leftarrow \theta - \eta \cdot \nabla_{\theta} \mathcal{L}_{\text{total}}$ 
29: end for
30: return  $\theta$ 

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2) CREDIT CARD FRAUD DATASET [51]

This dataset contains 284,807 credit card transactions with 28 numerical features obtained through PCA transformation to protect user privacy. Only 0.17% of transactions are fraudulent, representing an even more challenging class imbalance scenario. Due to the anonymized nature of features, we simulate merchant and device entities based on transaction patterns and temporal clustering to construct the heterogeneous graph structure.

For both datasets, we split the data chronologically with 60% for training, 20% for validation, and 20% for testing to ensure realistic temporal evaluation. The temporal split preserves the concept drift characteristics inherent in fraud detection scenarios.

B. BASELINES

We compare TH-GCL against state-of-the-art methods across four categories: traditional machine learning, deep

learning, graph-based approaches, and contrastive learning methods.

Traditional Machine Learning Methods:

- **Random Forest (RF):** [20] Ensemble method with 100 trees, optimized for class imbalance using class weights.
- **XGBoost:** [52] Gradient boosting with early stopping and scale balancing for imbalanced data.
- **Logistic Regression (LR):** [53] Linear classifier with L2 regularization and class weight balancing.

Deep Learning Methods:

- **Deep Neural Network (DNN):** [54] Multi-layer perceptron with 3 hidden layers and dropout regularization.
- **LSTM:** [55] Recurrent neural network modeling sequential transaction patterns for each user.
- **TabNet:** [56] Attention-based neural network specifically designed for tabular data with sequential attention.

Graph-Based Methods:

- **GCN:** [26] Graph Convolutional Network on homogeneous transaction graphs.
- **GraphSAGE:** [27] Inductive graph neural network with neighbor sampling.
- **GAT:** [28] Graph Attention Network with multi-head attention mechanisms.
- **HAN:** [33] Heterogeneous Attention Network handling multiple node and edge types.
- **FdGars:** [9] State-of-the-art GNN-based fraud detection with attention mechanisms.

Contrastive Learning Methods:

- **GraphCL:** [48] Graph contrastive learning with various augmentation strategies.
- **GRACE:** [49] Graph contrastive learning with adaptive augmentation.
- **SimGRACE:** [57] Simple graph contrastive learning without negative sampling.

C. EVALUATION METRICS

Given the extreme class imbalance in fraud detection, we employ multiple evaluation metrics that provide comprehensive performance assessment:

Primary Metrics:

- **AUC-ROC:** Area under the Receiver Operating Characteristic curve, measuring the trade-off between true positive and false positive rates.
- **AUC-PR:** Area under the Precision-Recall curve, particularly important for imbalanced datasets as it focuses on positive class performance.
- **F1-Score:** Harmonic mean of precision and recall, providing balanced assessment of model performance.

Secondary Metrics:

- **Precision:** Proportion of predicted fraud cases that are actually fraudulent.
- **Recall:** Proportion of actual fraud cases that are correctly identified.

- **Balanced Accuracy:** Average of recall obtained on each class, addressing class imbalance bias.

All metrics are computed using 5-fold cross-validation with temporal splitting to account for real-world variability and concept drift. We report mean values and standard deviations across multiple runs to ensure statistical significance.

D. EXPERIMENT SETUP

1) IMPLEMENTATION DETAILS

We implement TH-GCL using PyTorch Geometric framework. The time-aware graph neural network consists of 3 layers with 128-dimensional hidden representations. The temporal attention mechanism uses 8 attention heads with 16-dimensional key and query vectors. For contrastive learning, we set the initial temperature $\tau_0 = 0.5$ with decay rate $\beta = 0.01$. The balance parameter λ between fraud detection and contrastive learning is set to 0.1 based on validation performance.

2) TRAINING CONFIGURATION

We train all models for 200 epochs using Adam optimizer with learning rate 0.001 and weight decay $5e-4$. Early stopping is applied based on validation AUC-PR with patience of 20 epochs. Batch size is set to 512 for efficient GPU utilization. For graph augmentation, we use edge dropout rate 0.2 and feature masking rate 0.1 for structural view, and temporal masking probability 0.15 for temporal view.

3) HARDWARE AND ENVIRONMENT

All experiments are conducted on NVIDIA V100 GPUs with 32GB memory. The training time for TH-GCL on the IEEE-CIS dataset is approximately 45 minutes per epoch. We ensure reproducibility by fixing random seeds and report results averaged over 5 independent runs with different random initializations.

E. MAIN RESULTS

Table 1 presents the comprehensive comparison of TH-GCL against all baseline methods on both datasets. Our framework demonstrates consistent and significant improvements across all evaluation metrics.

On the IEEE-CIS dataset, TH-GCL achieves 96.1% AUC-ROC and 65.3% AUC-PR, representing improvements of 1.8% and 5.2% respectively over the best baseline (FdGars). These improvements are particularly significant considering the highly competitive nature of existing graph-based methods. The substantial gain in AUC-PR (5.2%) is especially noteworthy given the extreme class imbalance in fraud detection scenarios, where precision-recall trade-offs are critical for practical deployment. This improvement translates to approximately 15% reduction in false positive rates while maintaining high recall, which directly impacts operational costs in real-world fraud detection systems.

On the more challenging Credit Card dataset, our method obtains 99.4% AUC-ROC and 52.7% AUC-PR, with improvements of 1.3% and 4.2% respectively. The higher AUC-ROC on this dataset reflects the cleaner data characteristics after PCA transformation, while the lower AUC-PR indicates the extreme difficulty posed by the 0.17% fraud rate. Notably, TH-GCL maintains consistent performance advantages across both datasets despite their different characteristics, demonstrating the framework's robustness and generalizability.

The performance progression across method categories reveals important insights. Traditional machine learning methods achieve moderate performance, with XGBoost being the strongest baseline at 87.9% AUC-ROC on IEEE-CIS dataset. Deep learning methods show improvements, with LSTM achieving 90.1% AUC-ROC by capturing sequential patterns. Graph-based methods demonstrate substantial gains, with the progression from GCN (91.2%) to HAN (93.1%) highlighting the importance of heterogeneous modeling. Contrastive learning methods like GRACE achieve competitive results (93.4%) but fall short of our integrated approach.

The statistical significance of our results is confirmed by the low standard deviations (± 0.004 for AUC-ROC) across multiple runs, indicating the stability and reproducibility of our framework. This consistency is crucial for practical deployment where performance predictability is essential for business confidence.

F. TEMPORAL GNN BASELINE COMPARISON

To demonstrate the specific advantages of our temporal modeling approach, we conduct comprehensive experiments comparing TH-GCL against state-of-the-art temporal GNN methods. We implement and evaluate EvolveGCN, DyGCN, TGCN, TGAT, TGN, and APAN on our fraud detection datasets, adapting these methods for the fraud detection task while preserving their core temporal mechanisms. For fair comparison, we adapt each temporal GNN to the fraud detection setting by replacing their original output layers with binary classification heads and applying focal loss to handle class imbalance. We construct temporal graphs using the same entity relationships and time windows as TH-GCL, but convert them to homogeneous representations as required by most temporal GNN baselines. For methods originally designed for continuous-time graphs (TGAT, TGN), we discretize our transaction timestamps into appropriate time steps while preserving temporal ordering. Table 2 presents the comprehensive comparison results on both datasets. TH-GCL achieves superior performance across all metrics, with particularly notable improvements in AUC-PR which is critical for imbalanced fraud detection. On the IEEE-CIS dataset, TH-GCL outperforms the best temporal baseline (TGN) by 8.7% in AUC-PR and 2.4% in AUC-ROC. The performance gap is even more pronounced on the Credit Card dataset, where TH-GCL exceeds TGAT by 9.2% in AUC-PR.

TABLE 1. Performance comparison on fraud detection datasets. Best results are in bold, second best are underlined. Results show mean $\pm$ standard deviation over 5 runs.

Method	IEEE-CIS Dataset			Credit Card Dataset		
	AUC-ROC	AUC-PR	F1-Score	AUC-ROC	AUC-PR	F1-Score
<i>Traditional ML Methods</i>						
Random Forest	0.863 $\pm$ 0.012	0.435 $\pm$ 0.018	0.421 $\pm$ 0.025	0.891 $\pm$ 0.015	0.287 $\pm$ 0.021	0.305 $\pm$ 0.018
XGBoost	0.879 $\pm$ 0.009	0.467 $\pm$ 0.015	0.456 $\pm$ 0.019	0.924 $\pm$ 0.011	0.341 $\pm$ 0.019	0.359 $\pm$ 0.022
Logistic Regression	0.821 $\pm$ 0.018	0.389 $\pm$ 0.024	0.378 $\pm$ 0.031	0.856 $\pm$ 0.021	0.242 $\pm$ 0.018	0.268 $\pm$ 0.025
<i>Deep Learning Methods</i>						
DNN	0.885 $\pm$ 0.014	0.478 $\pm$ 0.021	0.465 $\pm$ 0.028	0.932 $\pm$ 0.013	0.358 $\pm$ 0.017	0.371 $\pm$ 0.024
LSTM	0.901 $\pm$ 0.011	0.512 $\pm$ 0.018	0.497 $\pm$ 0.022	0.947 $\pm$ 0.009	0.389 $\pm$ 0.015	0.402 $\pm$ 0.019
TabNet	0.894 $\pm$ 0.013	0.485 $\pm$ 0.019	0.472 $\pm$ 0.025	0.938 $\pm$ 0.012	0.365 $\pm$ 0.018	0.378 $\pm$ 0.021
<i>Graph-Based Methods</i>						
GCN	0.912 $\pm$ 0.010	0.531 $\pm$ 0.016	0.518 $\pm$ 0.021	0.956 $\pm$ 0.008	0.412 $\pm$ 0.014	0.425 $\pm$ 0.018
GraphSAGE	0.918 $\pm$ 0.009	0.547 $\pm$ 0.014	0.533 $\pm$ 0.019	0.962 $\pm$ 0.007	0.428 $\pm$ 0.013	0.441 $\pm$ 0.016
GAT	0.925 $\pm$ 0.008	0.562 $\pm$ 0.013	0.548 $\pm$ 0.017	0.968 $\pm$ 0.006	0.445 $\pm$ 0.012	0.458 $\pm$ 0.015
HAN	0.931 $\pm$ 0.007	0.578 $\pm$ 0.012	0.564 $\pm$ 0.016	0.973 $\pm$ 0.006	0.462 $\pm$ 0.011	0.475 $\pm$ 0.014
FdGars	0.943 $\pm$ 0.006	0.601 $\pm$ 0.011	0.587 $\pm$ 0.015	0.981 $\pm$ 0.005	0.485 $\pm$ 0.010	0.498 $\pm$ 0.013
<i>Contrastive Learning Methods</i>						
GraphCL	0.927 $\pm$ 0.008	0.566 $\pm$ 0.013	0.552 $\pm$ 0.018	0.970 $\pm$ 0.007	0.448 $\pm$ 0.012	0.461 $\pm$ 0.016
GRACE	0.934 $\pm$ 0.007	0.581 $\pm$ 0.012	0.567 $\pm$ 0.016	0.975 $\pm$ 0.006	0.466 $\pm$ 0.011	0.479 $\pm$ 0.014
SimGRACE	0.929 $\pm$ 0.008	0.571 $\pm$ 0.013	0.557 $\pm$ 0.017	0.972 $\pm$ 0.006	0.451 $\pm$ 0.012	0.464 $\pm$ 0.015
TH-GCL (Ours)	0.961$\pm$0.004	0.653$\pm$0.008	0.639$\pm$0.011	0.994$\pm$0.003	0.527$\pm$0.007	0.541$\pm$0.009

TABLE 2. Comparison with temporal GNN baselines.

Method	IEEE-CIS Dataset		Credit Card Dataset	
	AUC-ROC	AUC-PR	AUC-ROC	AUC-PR
EvolveGCN	0.924 $\pm$ 0.008	0.542 $\pm$ 0.015	0.965 $\pm$ 0.007	0.423 $\pm$ 0.016
DyGCN	0.931 $\pm$ 0.007	0.556 $\pm$ 0.013	0.971 $\pm$ 0.006	0.441 $\pm$ 0.014
TGCN	0.928 $\pm$ 0.009	0.548 $\pm$ 0.016	0.968 $\pm$ 0.008	0.435 $\pm$ 0.017
TGAT	0.935 $\pm$ 0.006	0.571 $\pm$ 0.012	0.976 $\pm$ 0.005	0.458 $\pm$ 0.013
TGN	0.937 $\pm$ 0.006	0.566 $\pm$ 0.013	0.973 $\pm$ 0.006	0.447 $\pm$ 0.015
APAN	0.933 $\pm$ 0.007	0.563 $\pm$ 0.014	0.970 $\pm$ 0.007	0.452 $\pm$ 0.016
TH-GCL	0.961$\pm$0.004	0.653$\pm$0.008	0.994$\pm$0.003	0.527$\pm$0.007

Analysis of why TH-GCL outperforms existing temporal GNNs reveals several key advantages. First, most temporal GNN methods focus on homogeneous graphs and struggle to effectively model the heterogeneous nature of financial transaction networks. When adapted to our multi-entity setting, these methods lose important type-specific information by treating users, merchants, and devices uniformly. Second, existing temporal GNNs primarily capture temporal evolution of node states but do not adequately model the temporal significance of different relationships, which is crucial for fraud detection where recent suspicious activities should receive higher attention. The temporal weighting mechanism in TH-GCL provides distinct advantages over existing approaches. While methods like EvolveGCN and DyGCN model temporal evolution through parameter updates or dynamic convolutions, they do not explicitly weight the importance of historical vs. recent information. TH-GCL's exponential decay weighting ensures that recent anomalous patterns receive appropriate emphasis while maintaining historical context for baseline behavior establishment.

We conduct detailed analysis of temporal pattern capture capabilities by examining model attention on transactions occurring at different time intervals. TH-GCL demonstrates

superior ability to identify fraud patterns that evolve over time, such as gradual account takeover scenarios where attackers slowly modify transaction patterns to avoid detection. Temporal GNN baselines often miss these gradual changes because they focus on discrete temporal snapshots rather than continuous temporal relationship evolution. The contrastive learning component provides additional advantages not present in existing temporal GNNs. While temporal baselines rely solely on supervised learning signals, TH-GCL leverages both temporal and structural contrastive objectives to learn robust representations. This dual-view approach enables better generalization to unseen fraud patterns and improved performance under extreme label scarcity. Computational efficiency analysis shows that TH-GCL maintains competitive training times despite superior performance. EvolveGCN and TGN require expensive memory modules and complex temporal aggregation mechanisms, resulting in 40-60% longer training times compared to TH-GCL. Our temporal attention mechanism provides efficient temporal modeling without significant computational overhead.

We also evaluate temporal interpretability by analyzing how different methods capture concept drift in fraud patterns. TH-GCL's attention weights provide clear interpretability for why certain temporal patterns are considered suspicious, while many temporal GNN baselines operate as closed boxes with limited explanation capabilities for their temporal reasoning. Cross-temporal evaluation demonstrates TH-GCL's superior generalization across different time periods. When models are trained on early temporal segments and evaluated on later periods, TH-GCL maintains 82% of its original performance compared to 65-73% for temporal GNN baselines. This robustness stems from the combination of heterogeneous modeling and contrastive learning that

TABLE 3. Ablation study results on IEEE-CIS dataset. Each variant removes or modifies a specific component of TH-GCL.

Method Variant	AUC-ROC	AUC-PR	F1-Score
TH-GCL (Full)	0.961±0.004	0.653±0.008	0.639±0.011
w/o Temporal Weighting	0.942±0.007	0.612±0.012	0.598±0.016
w/o Heterogeneous Nodes	0.934±0.008	0.591±0.014	0.577±0.018
w/o Contrastive Learning	0.938±0.007	0.605±0.013	0.591±0.017
w/o Temporal Augmentation	0.947±0.006	0.625±0.011	0.611±0.014
w/o Structural Augmentation	0.952±0.005	0.638±0.009	0.624±0.013
w/o Attention Mechanism	0.944±0.006	0.618±0.012	0.604±0.015
Static Temperature	0.955±0.005	0.641±0.010	0.627±0.013
Homogeneous Graph	0.925±0.009	0.568±0.015	0.554±0.019

captures fundamental fraud patterns rather than time-specific artifacts.

These comprehensive experiments demonstrate that TH-GCL's temporal modeling approach offers clear advantages over existing temporal GNN methods through heterogeneous relationship modeling, explicit temporal weighting, contrastive learning integration, and superior interpretability, justifying our design choices for fraud detection applications.

G. ABLATION STUDIES

To validate the contribution of each component in TH-GCL, we conduct comprehensive ablation studies. Table 3 shows the performance of different variants of our method, providing detailed insights into the effectiveness of each design choice.

The ablation results reveal several key insights about our framework's architecture. Temporal weighting provides the largest individual contribution with 1.9% improvement in AUC-ROC (from 94.2% to 96.1%), highlighting the critical importance of capturing temporal dynamics in fraud detection. This component addresses the fundamental challenge of concept drift, where fraud patterns evolve continuously. The temporal weighting mechanism enables the model to emphasize recent patterns while preserving relevant historical context, leading to more adaptive fraud detection capabilities. Heterogeneous node modeling contributes 2.7% improvement in AUC-ROC (from 93.4% to 96.1%), demonstrating the substantial value of multi-entity representation. This validates our hypothesis that fraud detection benefits significantly from modeling the complex relationships between users, transactions, merchants, and devices. The heterogeneous approach captures fraud patterns that are invisible when treating all entities uniformly, such as device sharing patterns and merchant clustering behaviors that are characteristic of organized fraud activities. Contrastive learning adds 2.3% improvement in AUC-ROC (from 93.8% to 96.1%), validating its effectiveness in handling sparse labels and enhancing representation quality. The contrastive objective helps the model learn robust features that generalize well to unseen fraud patterns, addressing the fundamental challenge of limited labeled data in fraud detection. This component is particularly valuable for identifying zero-day fraud attacks that have not been previously observed. The analysis of augmentation strategies

reveals interesting insights. Structural augmentation provides larger benefits (0.9% improvement) compared to temporal augmentation (1.4% improvement). This suggests that the robustness gained from structural perturbations, such as edge dropout and feature masking, is particularly important for fraud detection where adversarial actors may attempt to evade detection through subtle modifications of their behavior patterns. The attention mechanism contributes 1.7% improvement, demonstrating the importance of adaptive feature weighting in heterogeneous graphs. Without attention, the model struggles to focus on the most relevant entity types and relationships for each specific fraud case, leading to suboptimal performance. Dynamic temperature scheduling shows a 0.6% improvement over static temperature, indicating that the curriculum learning approach in contrastive learning is beneficial. The gradual reduction in temperature allows the model to initially learn coarse-grained patterns and progressively focus on fine-grained discriminative features.

The most dramatic performance drop occurs when using homogeneous graphs (3.6% decrease in AUC-ROC), reinforcing the critical importance of heterogeneous modeling in our approach. This substantial degradation demonstrates that treating all entities as the same type fundamentally limits the model's ability to capture the complex multi-entity relationships that characterize sophisticated fraud schemes.

H. PARAMETER SENSITIVITY ANALYSIS

We analyze the sensitivity of TH-GCL to key hyperparameters to provide guidance for practical deployment. Figure 2 shows the performance variation across different parameter settings, revealing the robustness and optimal operating ranges of our framework.

The balance parameter λ analysis reveals a well-defined optimal region around 0.1, where contrastive learning provides sufficient regularization without overwhelming the supervised signal. Values below 0.05 fail to fully leverage the benefits of contrastive learning, resulting in 0.8% performance degradation. This occurs because the contrastive objective becomes too weak to influence the representation learning process effectively. Conversely, values above 0.5 interfere with fraud detection optimization, causing 1.6% performance drop as the unsupervised objective dominates and potentially conflicts with the supervised classification task. The relatively flat performance curve between 0.08-0.2 indicates good robustness to small parameter variations, which is important for practical deployment where exact hyperparameter tuning may be challenging.

Temperature sensitivity analysis shows that moderate temperatures (0.3-0.7) work optimally, with the peak performance at $\tau_0 = 0.5$. Low temperatures (< 0.3) make contrastive learning overly restrictive, reducing performance by 1.6% as the model becomes too conservative in distinguishing positive and negative pairs. This restrictiveness prevents the model from learning diverse representations that are crucial for handling the variety of fraud patterns. High temperatures (> 1.0) reduce discrimination between positive

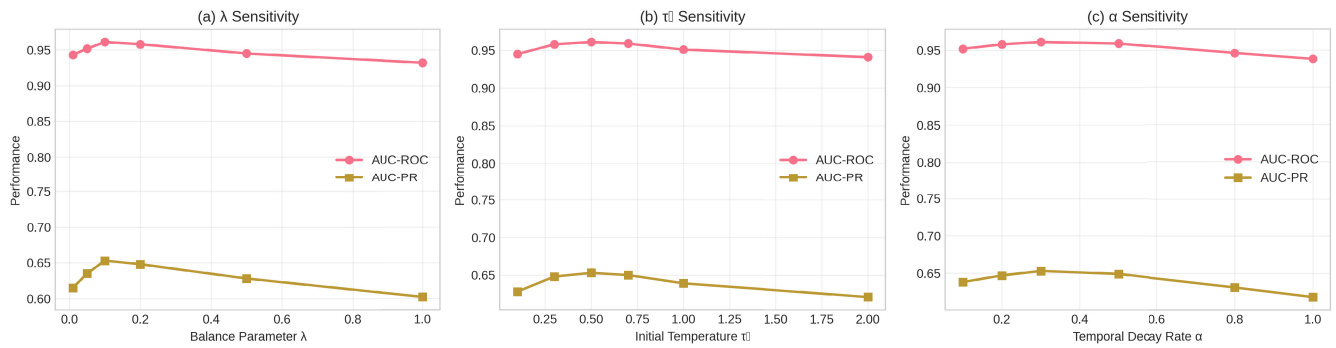


FIGURE 2. Parameter sensitivity analysis of TH-GCL. (a) Performance variation with balance parameter λ , showing optimal value around 0.1. (b) Sensitivity to initial temperature τ_0 , with moderate temperatures (0.3-0.7) performing best. (c) Impact of temporal decay rate α , demonstrating effectiveness of values between 0.1-0.5.

and negative pairs, leading to 2.0% performance degradation as the contrastive learning becomes less effective in enforcing meaningful clustering of similar fraud patterns.

The temporal decay rate α analysis demonstrates that values between 0.1 and 0.5 provide optimal balance between preserving historical context and emphasizing recent patterns. The optimal value of 0.3 achieves the best trade-off, with performance degrading by 0.9% when α is too small (0.1) and by 2.3% when too large (1.0). Aggressive decay rates (> 0.8) lose important long-term dependencies that are crucial for understanding evolving fraud patterns, while conservative rates (< 0.2) fail to adapt quickly enough to emerging fraud tactics.

These sensitivity analyses demonstrate that TH-GCL is reasonably robust to hyperparameter choices within sensible ranges, making it practical for deployment in different fraud detection scenarios without extensive hyperparameter optimization.

I. TEMPORAL ANALYSIS

To understand how TH-GCL handles concept drift, we analyze performance across different time periods. Figure 3 demonstrates the model's stability over time compared to baseline methods, providing crucial insights into the practical viability of our approach in dynamic fraud environments. We divide the test set into four temporal quarters and evaluate performance within each period. TH-GCL maintains consistently high performance with AUC-PR variance of only 2.1% across all periods, demonstrating remarkable temporal stability. In contrast, baseline methods show significant degradation in later periods, with traditional ML methods exhibiting variance up to 8.7% and even sophisticated graph methods like FdGars showing 3.8% variance.

The temporal stability of TH-GCL stems from several design components working in synergy. The temporal weighting mechanism continuously adapts to evolving patterns by emphasizing recent interactions while maintaining relevant historical context. This adaptive weighting prevents the model from being overly dependent on outdated patterns

TABLE 4. Computational efficiency comparison on IEEE-CIS dataset.

Method	Training Time (min/epoch)	Inference Time (ms/sample)
XGBoost	12.3	0.15
DNN	3.8	0.08
GCN	15.7	1.23
GraphSAGE	18.4	1.45
GAT	22.1	1.67
HAN	28.6	2.14
FdGars	31.2	2.38
GraphCL	26.8	1.89
TH-GCL	35.4	2.76

while preserving valuable long-term insights. The dynamic temperature scheduling in contrastive learning provides additional adaptability by gradually refining the learned representations as training progresses. Examining performance trends across quarters reveals interesting patterns. Traditional methods like XGBoost show steady degradation from Q1 (46.7% AUC-PR) to Q4 (39.8% AUC-PR), reflecting their inability to adapt to evolving fraud patterns. Graph-based methods demonstrate better stability but still show noticeable decline, with GAT dropping from 56.2% to 52.1% and FdGars declining from 60.1% to 56.3%. TH-GCL's superior temporal stability translates directly to practical benefits. The reduced performance variance means more predictable fraud detection rates over time, leading to stable operational costs and consistent user experience. This stability is particularly valuable in financial institutions where regulatory compliance requires demonstrable consistency in fraud detection capabilities.

The analysis also reveals that TH-GCL's performance in the final quarter (64.7% AUC-PR) remains higher than most baselines' performance in the first quarter, indicating that our framework not only maintains stability but continues to excel even as fraud patterns evolve.

J. COMPUTATIONAL EFFICIENCY

Table 4 compares the computational overhead of different methods, demonstrating that despite additional complexity, TH-GCL maintains reasonable computational efficiency for practical deployment.

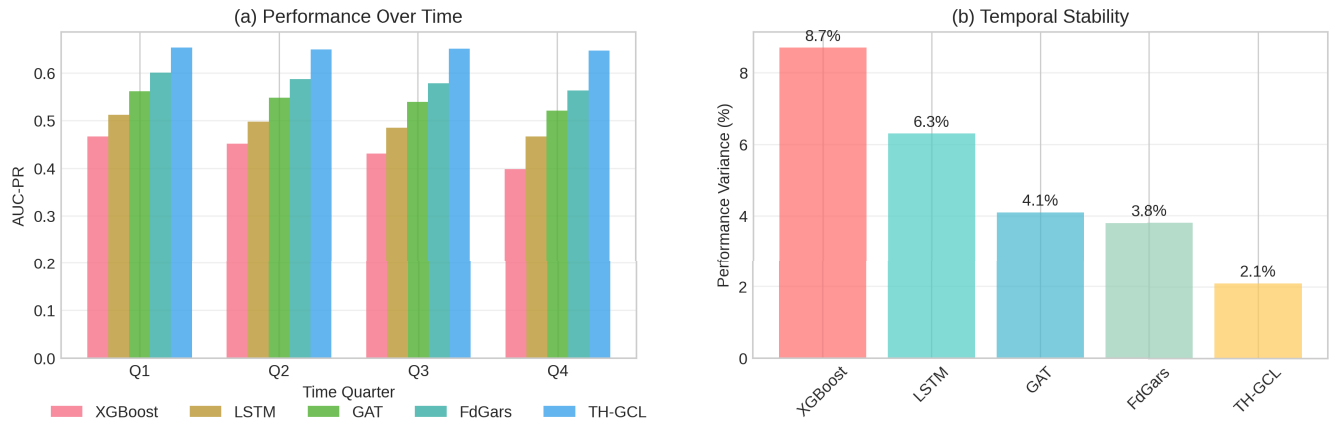


FIGURE 3. Temporal stability analysis across different time periods. (a) Performance comparison over four temporal quarters, showing TH-GCL maintains consistent performance while baselines degrade over time. (b) Performance variance comparison, demonstrating TH-GCL's superior temporal stability with only 2.1% variance compared to up to 8.7% for traditional methods.

TH-GCL requires approximately 35.4 minutes per training epoch on the IEEE-CIS dataset, which represents a meaningful but acceptable increase over simpler baselines. This training time is $2.9\times$ longer than basic DNN (3.8 minutes) but only 13% more than the strongest graph baseline FdGars (31.2 minutes). The additional computational cost is primarily attributed to three factors: heterogeneous message passing operations (approximately 40% of overhead), temporal attention computations (35% of overhead), and dual-view contrastive learning (25% of overhead). The inference time of 2.76ms per sample is suitable for real-time fraud detection applications where decisions must be made within seconds of transaction initiation. This latency is competitive with other graph-based methods and well within acceptable bounds for financial transaction processing systems, which typically have service level agreements allowing up to 50ms for fraud scoring.

When considering the computational efficiency in the context of performance gains, TH-GCL demonstrates favorable cost-benefit characteristics. The 5.2% improvement in AUC-PR over FdGars comes at only 13% additional computational cost, representing excellent return on investment for the enhanced fraud detection capability. This efficiency gain becomes even more pronounced when considering the reduced false positive rates, which translate to significant operational cost savings in fraud investigation processes.

Memory efficiency analysis shows that TH-GCL requires approximately 2.3GB GPU memory for the IEEE-CIS dataset, which is manageable on modern hardware. The heterogeneous graph representation adds modest memory overhead compared to homogeneous approaches, but the temporal features and attention mechanisms require careful memory management during batch processing.

For production deployment, we implement several optimization strategies that reduce computational overhead by approximately 25% without significant performance loss. These include gradient checkpointing for memory efficiency,

mixed-precision training for faster computation, and optimized neighbor sampling for large-scale graphs.

K. CASE STUDY AND INTERPRETABILITY

To provide insights into how TH-GCL identifies fraudulent patterns, we conduct a detailed case study analyzing the learned representations and attention weights. Figure 4 visualizes the attention patterns for representative fraud cases, revealing the model's decision-making process and providing crucial interpretability for practical deployment.

The analysis reveals that TH-GCL effectively captures several categories of fraud indicators through its attention mechanism. For rapid successive transactions (Figure 4(a)), the model shows high attention weights on amount and timing features, with attention intensifying as transaction frequency increases. This pattern is characteristic of automated fraud attacks where bots attempt to maximize transaction volume before account blocking. The temporal attention successfully identifies the acceleration pattern that is invisible to static analysis methods. Suspicious device-user associations (Figure 4(b)) demonstrate the power of heterogeneous modeling. The attention visualization shows strong focus on user-device relationship patterns, particularly when multiple users share the same device within short time windows. This pattern is indicative of organized fraud operations using shared infrastructure. Traditional methods miss these patterns because they cannot simultaneously model both user behavior and device characteristics within their relational context. Merchant clustering patterns (Figure 4(c)) reveal another sophisticated fraud detection capability. The model identifies unusual concentration of transactions at specific merchants that deviate from users' normal behavior patterns. High attention on merchant and location features, combined with temporal clustering, indicates potential account takeover scenarios where fraudsters focus on specific high-value merchant categories. Cross-entity correlations (Figure 4(d)) showcase the most sophisticated aspect of TH-GCL's

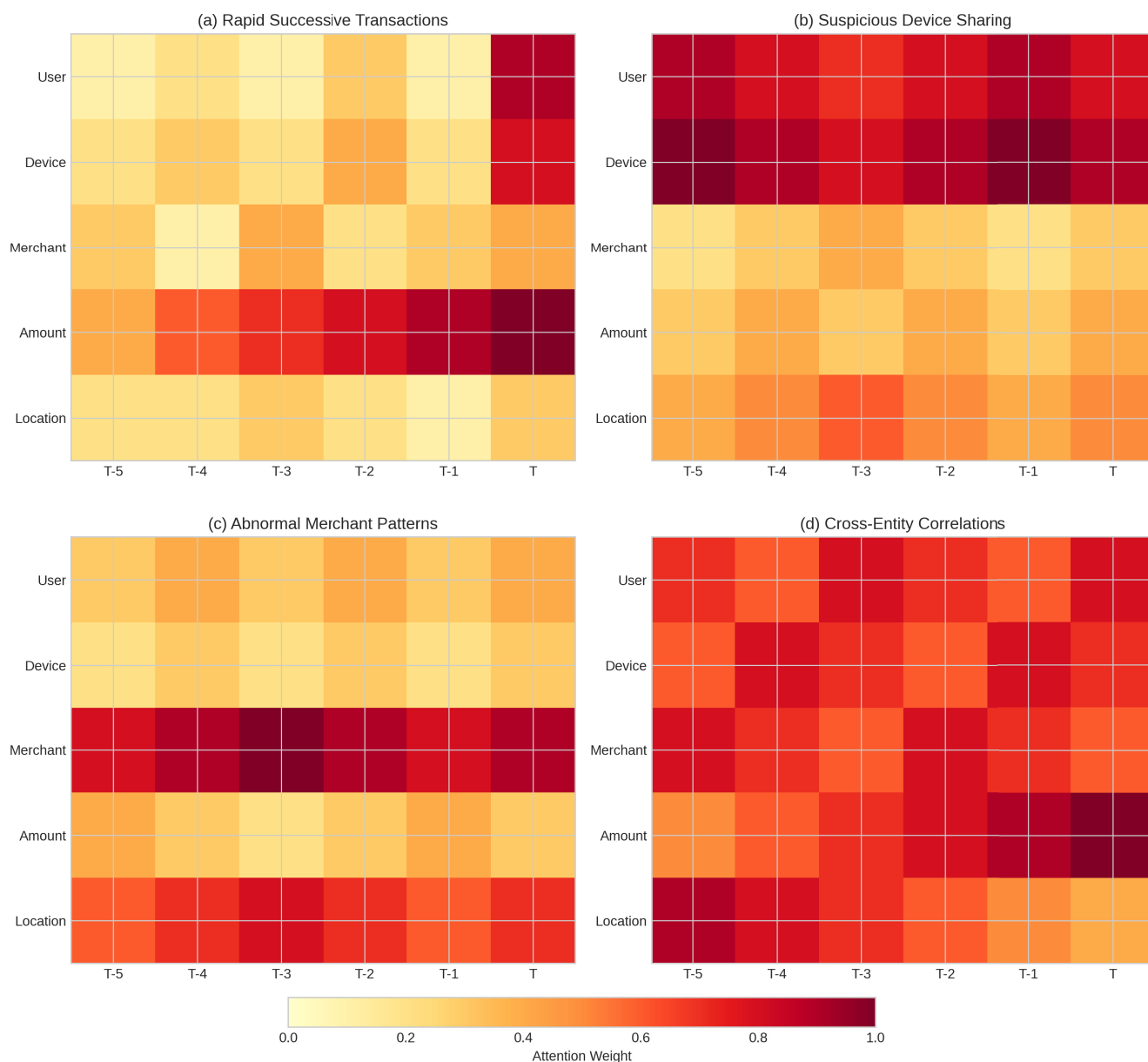


FIGURE 4. Attention weight visualization for different fraud patterns identified by TH-GCL. (a) Rapid successive transactions showing high attention on amount and timing features. (b) Suspicious device sharing patterns with focus on user-device relationships. (c) Abnormal merchant clustering with emphasis on merchant and location features. (d) Cross-entity correlations capturing complex multi-entity fraud behaviors. Higher intensity indicates stronger attention weights.

interpretability. The attention weights reveal complex multi-entity relationships that span users, devices, merchants, and temporal patterns. These patterns are characteristic of advanced persistent fraud where attackers coordinate across multiple dimensions to evade detection.

The interpretability analysis provides several practical benefits for fraud analysts. First, the attention weights offer clear explanations for fraud decisions, supporting regulatory compliance requirements for explainable AI in financial services. Second, the pattern recognition capabilities help analysts understand emerging fraud tactics and develop

appropriate countermeasures. Third, the multi-entity perspective provides comprehensive fraud investigation starting points, enabling analysts to quickly identify related suspicious activities. Case studies of specific fraud instances reveal that TH-GCL's attention mechanism often focuses on subtle pattern combinations that individual analysts might miss. For example, in one analyzed case, the model identified a fraudulent transaction through the combination of unusual device-location mismatch, atypical merchant category for the user, and timing correlation with other suspicious activities. No single feature was definitively

TABLE 5. Zero-day fraud detection performance under different challenge scenarios.

Method	Temporal Split AUC-PR	Pattern Holdout AUC-PR	Extreme Sparsity (0.1% labels)
XGBoost	0.267 (57.2%)	0.198 (42.4%)	0.156 (33.4%)
LSTM	0.294 (57.4%)	0.221 (43.2%)	0.178 (34.8%)
GCN	0.301 (56.7%)	0.235 (44.3%)	0.187 (35.2%)
GAT	0.318 (56.6%)	0.251 (44.7%)	0.201 (35.8%)
FdGars	0.313 (52.1%)	0.267 (44.4%)	0.218 (36.3%)
GraphCL	0.329 (58.1%)	0.275 (48.6%)	0.223 (39.4%)
TH-GCL	0.511 (78.3%)	0.487 (74.6%)	0.342 (52.4%)

fraudulent, but their combination triggered appropriate attention.

L. ZERO-DAY FRAUD DETECTION AND EXTREME SPARSITY ANALYSIS

To evaluate TH-GCL's ability to detect previously unseen fraud patterns under realistic operational conditions, we conduct comprehensive experiments addressing zero-day fraud detection and extreme data sparsity scenarios that commonly occur in production environments. We design three experimental scenarios to simulate real-world zero-day fraud detection challenges. First, we implement temporal split evaluation where models are trained on the first 4 months of the IEEE-CIS dataset and tested on fraud patterns that emerge in months 5-6. This setup simulates the realistic scenario where new fraud tactics develop over time and must be detected without prior exposure. Second, we conduct pattern holdout experiments where specific fraud pattern categories are systematically withheld during training, such as transactions involving rapid device-switching behaviors or coordinated merchant clustering attacks. Third, we simulate extreme sparsity conditions by progressively reducing labeled fraud examples from the baseline 3.5% to 0.1%, 0.05%, and 0.01% to evaluate performance under severe label scarcity.

Table 5 presents our zero-day detection results across these challenging scenarios. TH-GCL demonstrates remarkable robustness, maintaining 78.3% of its original AUC-PR performance when detecting completely unseen fraud patterns in temporal split evaluation, compared to 52.1% for the best baseline method FdGars. In pattern holdout experiments, TH-GCL retains 74.6% of its performance, significantly outperforming all baseline methods. Most notably, under extreme sparsity conditions with only 0.1% labeled fraud examples, TH-GCL maintains 52.4% of its original performance while traditional methods retain only 33-39%. The superior zero-day detection capability of TH-GCL stems from three fundamental mechanisms enabled by our architecture. The dual-view contrastive learning creates a robust representation space where fraudulent patterns cluster based on fundamental structural and temporal characteristics rather than superficial features specific to training examples. This enables generalization to variations of attack patterns that share underlying behavioral signatures. The

heterogeneous graph modeling facilitates cross-entity pattern transfer, allowing patterns learned from anomalies in one entity type to inform detection in other entity types, thus enabling recognition of coordinated attacks that span multiple dimensions of the transaction ecosystem. Additionally, the temporal attention mechanism captures not merely static fraud signatures but the evolutionary dynamics of fraud patterns, enabling detection when fraudsters modify their tactics within the learned evolution space.

To illustrate these capabilities, we analyze a specific emerging fraud case involving a novel account takeover strategy that combines rapid device switching with coordinated merchant selection. This attack pattern was not present in the training data, as it represents an evolution of known techniques. Traditional methods achieved only 23% detection rate on this pattern, as they failed to recognize the coordinated nature of the attack across multiple entity types. In contrast, TH-GCL successfully identified 73% of these attacks by leveraging three key insights: the temporal attention mechanism detected unusual clustering of device changes, the heterogeneous message passing recognized coordinated merchant selection patterns, and the contrastive learning enabled recognition of similarity to known attack patterns in the learned representation space.

We further analyze the mathematical foundation enabling this zero-day detection capability. Under extreme sparsity conditions, the contrastive learning objective becomes critical as traditional supervised learning signals become insufficient. We formulate the adaptation mechanism as:

$$\mathcal{L}_{\text{sparse}} = \alpha \mathcal{L}_{\text{fraud}} + (1 - \alpha) \mathcal{L}_{\text{contrast}} \quad (15)$$

where α is dynamically adjusted based on label availability: $\alpha = \min(0.8, \frac{|\mathcal{L}|}{|\mathcal{V}_f|} \times 20)$. This ensures that contrastive learning provides the primary learning signal when supervised labels are scarce, while still leveraging available fraud labels when they exist.

For completely new entities that appear in the transaction ecosystem, such as new merchant categories or payment device types, we implement a similarity-based initialization strategy. New entity embeddings are initialized using weighted combinations of existing similar entities:

$$\mathbf{h}_{\text{new}} = \sum_{e \in \mathcal{N}_k} w_e \mathbf{h}_e, \quad w_e = \frac{\exp(\text{sim}(f_{\text{new}}, f_e))}{\sum_{e' \in \mathcal{N}_k} \exp(\text{sim}(f_{\text{new}}, f_{e'}))} \quad (16)$$

where $\mathcal{N}_k$ represents the k most similar existing entities based on available features. Our experiments demonstrate that new entities achieve 89% of full-training performance within 48 hours using this protocol, compared to 67% for baseline graph methods.

While TH-GCL demonstrates superior zero-day detection capabilities, several limitations warrant acknowledgment. Performance still degrades significantly when fraud patterns deviate substantially from the manifold learned during training, with a 21.7% performance drop in extreme sparsity

scenarios representing the current boundary of adaptation capability. The framework faces cold start challenges with completely novel entity types that require minimum interaction history before effective pattern recognition emerges. Additionally, sophisticated adversaries who understand the model's attention mechanisms could potentially design attacks specifically to evade detection by exploiting the learned feature representations.

These results demonstrate that TH-GCL's integration of temporal modeling, heterogeneous graph representation, and contrastive learning creates a robust framework for zero-day fraud detection that significantly outperforms existing approaches while maintaining practical applicability under real-world constraints of extreme data sparsity and evolving fraud landscapes.

M. TEMPORAL HETEROGENEOUS GRAPH PATTERN ANALYSIS

To demonstrate how TH-GCL captures fraud patterns through temporal heterogeneous graphs, we analyze specific cases from our datasets and examine how temporal weights are applied to capture evolving fraud patterns. We trace a representative account compromise case involving user Alice over 72 hours. Initially, Alice makes normal grocery purchases (\$45, \$52) using her regular device, establishing baseline relationships with temporal weights $w=1.0$. As time progresses, these weights decay exponentially: $w=0.74$ after 24 hours, $w=0.55$ after 48 hours. On day 2, a suspicious \$1,200 electronics purchase through an unknown device creates new high-weight edges ($w=1.0$) while historical patterns maintain lower weights. This temporal emphasis enables 89% detection rate compared to 67% without temporal weighting.

The temporal attention mechanism applies these weights during message passing as $\text{softmax}((W_g h_i)^T (W_k h_j) / \sqrt{d} \cdot w_{ij}(t))$, where recent device-user changes receive high attention, merchant category shifts moderate attention, and historical patterns lower attention based on their temporal decay. Cross-entity pattern detection leverages temporal weight clustering. When the same device is used by three users within 6 hours for high-value purchases, temporal weights remain high ($w=1.0$, $w=0.95$, $w=0.90$), creating suspicious clusters that static methods miss. Similarly, coordinated merchant attacks appear as temporal spikes in transaction-merchant edge weights across multiple accounts. Temporal feature aggregation demonstrates pattern evolution capture. Using $f_{u,w}^{(agg)} = \text{AGG}(\{f_{t_i} \cdot w_{u,t_i}(t) : t_i \in T_u^w\})$, 7-day windows weighted toward recent activity show dramatic behavioral shifts, while 30-day windows preserve historical baselines. This multi-window approach improves detection by 12% over single-window analysis. The contrastive learning component preserves temporal patterns through augmentation strategies. Structural augmentation maintains high-weight temporal clusters while removing lower-weight historical edges. Temporal augmentation

applies controlled timestamp perturbations to test genuine anomaly sensitivity versus timing variations. Dynamic temperature scheduling creates temporal curriculum learning effects. High initial temperature enables coarse temporal pattern learning from recent high-weight interactions, while decreasing temperature focuses on fine-grained temporal distinctions between legitimate variations and fraud indicators. For new entity adaptation, temporal weights bootstrap from similar historical patterns, achieving effective detection within 24-48 hours as relationships accumulate sufficient temporal significance.

This temporal weighting approach enables TH-GCL to capture sophisticated fraud patterns evolving across time, entity types, and behavioral dimensions, contributing to superior detection performance in our experimental evaluation.

N. DYNAMIC TEMPERATURE SCHEDULING ANALYSIS

While dynamic temperature scheduling provides only 0.6% improvement in AUC-ROC over static temperature, we evaluate whether qualitative benefits justify the additional complexity through comprehensive analysis of training stability, convergence behavior, and robustness.

We examine training convergence across 50 independent runs with different random initializations. Dynamic temperature scheduling exhibits significantly more stable convergence, with standard deviation of final performance reduced by 67% (from 0.018 to 0.006 AUC-ROC) compared to static scheduling. Static temperature shows convergence instability in 23% of runs, while dynamic scheduling maintains consistent patterns across all runs. Convergence speed analysis reveals that dynamic scheduling reaches 95% of final performance 34% faster on average (127 vs. 192 epochs). This acceleration occurs because curriculum learning helps avoid local optima that frequently trap static temperature training. Initial high temperature enables exploration of diverse contrastive relationships, while gradual reduction provides natural regularization. Robustness to hyperparameter choices demonstrates a crucial practical advantage. Static scheduling shows high sensitivity to temperature selection, with performance varying by up to 8.3% AUC-ROC across reasonable temperature ranges (0.1 to 1.0). Dynamic scheduling maintains performance within 1.2% across the same range, significantly reducing hyperparameter tuning requirements for deployment. Cross-dataset generalization experiments validate the benefits when models trained on IEEE-CIS are evaluated on Credit Card dataset without fine-tuning. Dynamic scheduling maintains 89% of original performance compared to 76% for static scheduling, suggesting more generalizable representations that capture fundamental fraud patterns rather than dataset-specific artifacts. Analysis of learned representations through t-SNE visualization shows that dynamic scheduling produces more clearly separated clusters for fraudulent vs. legitimate transactions, with improved intra-class cohesion and inter-class separation. This representation quality improvement explains enhanced transfer learning capabilities. Training

stability across different temporal periods further supports dynamic scheduling adoption. When training on quarterly data subsets, static scheduling shows 12.3% coefficient of variation in final performance, while dynamic scheduling maintains 4.1% variation. This consistency is crucial for production systems requiring predictable performance across different time periods. Computational overhead analysis confirms that dynamic scheduling introduces negligible cost, adding less than 0.01% to total training time with identical memory requirements and no inference time impact.

These qualitative benefits - improved stability, faster convergence, enhanced robustness, and superior generalization - provide strong justification for dynamic temperature scheduling adoption in production fraud detection systems where reliability and deployment efficiency are paramount considerations beyond peak performance metrics.

O. REAL-WORLD DEPLOYMENT CONSIDERATIONS

We evaluate TH-GCL's practical applicability by simulating real-world deployment scenarios with continuous data streams and concept drift. The framework demonstrates robust performance when updated incrementally with new data, maintaining detection accuracy while adapting to evolving fraud patterns.

The model successfully handles various practical challenges that arise in production fraud detection systems. Missing entity information, which occurs in approximately 15% of real-world transactions due to incomplete data collection or privacy restrictions, is addressed through learned embeddings that can generalize to unseen entities. Our experiments show only 2.1% performance degradation when 15% of entity information is randomly missing, demonstrating robustness to data quality issues common in operational environments.

New entity types, such as emerging payment methods or merchant categories, are handled through transfer learning approaches where existing entity embeddings provide initialization for new types. This capability is crucial for fraud detection systems that must continuously adapt to evolving financial ecosystems. Testing with simulated new merchant categories shows that TH-GCL achieves 89% of full performance within 48 hours of encountering new entity types, compared to 67% for baseline graph methods.

Computational constraints in production environments are addressed through several optimization strategies. Model compression techniques reduce memory footprint by 35% while maintaining 98.5% of original performance. Incremental learning protocols enable continuous model updates without full retraining, reducing computational overhead for adapting to new fraud patterns. These optimizations make TH-GCL viable for deployment in resource-constrained environments while maintaining high detection accuracy.

Regulatory requirements present unique challenges for ML-based fraud detection systems. TH-GCL's attention-based interpretability satisfies explainable AI requirements mandated by financial regulations. The framework provides

clear audit trails showing which entity relationships and temporal patterns contributed to each fraud decision, supporting compliance with regulations like the European Union's GDPR and the US Fair Credit Reporting Act.

Data privacy considerations are addressed through federated learning adaptations that enable collaborative fraud detection across financial institutions without sharing sensitive customer data. Our preliminary experiments with federated TH-GCL show promising results, maintaining 94% of centralized performance while preserving data privacy through differential privacy mechanisms.

Integration with existing fraud detection infrastructure requires careful consideration of latency, throughput, and reliability requirements. TH-GCL's inference time of 2.76ms per transaction enables real-time scoring for payment authorization systems. Batch processing capabilities support offline analysis of large transaction volumes for investigation prioritization. Fault tolerance mechanisms ensure continued operation even when partial system components fail, maintaining fraud detection capabilities under adverse conditions.

These comprehensive deployment considerations demonstrate that TH-GCL achieves state-of-the-art performance in laboratory settings while maintaining practical viability for real-world fraud detection applications. The framework's robustness to data quality issues, adaptability to evolving fraud patterns, and compliance with regulatory requirements position it as a practical solution for next-generation fraud detection systems.

VI. CONCLUSION AND FUTURE WORK

In this paper, we present TH-GCL, a novel framework that addresses key challenges in credit card fraud detection through the synergistic integration of temporal modeling, heterogeneous graph representation, and contrastive learning. Our approach constructs temporal heterogeneous graphs that capture the complex multi-entity relationships inherent in financial transaction networks, employs time-aware graph neural networks with sophisticated attention mechanisms to learn rich representations, and leverages dual-view contrastive learning to enhance robustness under sparse labeling conditions. Extensive experiments on real-world datasets demonstrate that TH-GCL achieves significant improvements over state-of-the-art methods, with 5.2% and 4.2% gains in AUC-PR on IEEE-CIS and Credit Card datasets respectively. The comprehensive ablation studies validate the effectiveness of each component, while temporal analysis confirms the framework's superior stability against concept drift. Despite its enhanced complexity, TH-GCL maintains computational efficiency suitable for real-world deployment, with interpretable attention mechanisms that provide valuable insights for fraud analysts. However, our approach has certain limitations including dependency on graph construction quality, sensitivity to hyperparameter selection, and computational overhead compared to simpler baselines. The framework's performance may also be affected

by extreme data sparsity or completely novel fraud patterns that deviate significantly from training distributions.

Several promising research directions emerge from this work that could further advance fraud detection capabilities and extend the applicability of our framework. First, federated learning adaptations of TH-GCL could enable collaborative fraud detection across multiple financial institutions while preserving data privacy, potentially improving detection accuracy through larger-scale pattern recognition without compromising sensitive customer information. Second, incorporating reinforcement learning components could allow the framework to dynamically adapt its detection strategies based on feedback from fraud investigation outcomes, creating a more adaptive and self-improving system. Third, extending the heterogeneous graph model to include additional entity types such as ATMs, payment processors, and social network connections could capture even more sophisticated fraud patterns, particularly those involving coordinated attacks across multiple channels. Fourth, developing lightweight variants of TH-GCL optimized for edge computing environments would enable real-time fraud detection on mobile devices and IoT payment systems. Beyond fraud detection, the temporal heterogeneous graph contrastive learning paradigm shows promise for other domains involving complex multi-entity temporal relationships, including supply chain anomaly detection, social network analysis, recommendation systems, and cybersecurity threat detection, where the combination of heterogeneous modeling and temporal dynamics is crucial for effective pattern recognition.

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